Shape and Space 4 — Sine and cosine ratios · Calculating sides and angles

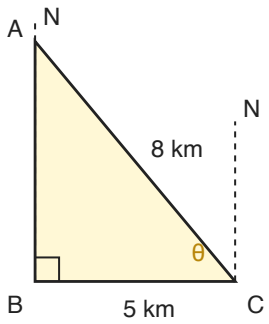
Calculator allowed. Give answers correct to 3 significant figures (angles to 1 decimal place where asked). Draw and label a right-angled triangle for each part. Marks in brackets. ★ = harder question.

1 A window cleaner's ladder is 6 m long and makes an angle of $70°$ with the level ground as it leans against a vertical wall. [4]

a Calculate the height of the top of the ladder above the ground.

b The cleaner climbs 2.5 m up the ladder. How far is his lower foot from the wall?

2 Three phone masts A, B and C: C is 5 km due east of B; A is due north of B and 8 km from C. [8]



a Calculate, to 1 decimal place: **(i)** the angle θ at C, **(ii)** the bearing of A from C, **(iii)** the bearing of C from A.

b Calculate the distance AB in km.

3 A circle with centre O represents the Earth (radius $OW = 6400$ km). A satellite S is at a distance $OS = 42\,000$ km from the centre. W (Washington) and D (Delhi) are points on the circle such that the line OS crosses the chord WD at right angles at N, and angle $WOD = 100°$. [13]

a Calculate, to 3 s.f. or to the nearest degree: **(i)** WN, **(ii)** ON, **(iii)** NS, **(iv)** angle ODS.

b Explain briefly why, if angle ODS were less than $90°$, a signal from the satellite could not reach Delhi.

c A radio signal is sent from W to D via S. Calculate the distance it travels.

d Radio signals travel at $300\,000$ km/s. How long does the signal take?

Total: 25 marks

Check: SOH CAH TOA — pick the ratio that uses the side you know and the side you want · hypotenuse is always opposite the right angle · angles from two sides: use $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$ · calculator in degrees, keep full values between steps.